GenreMaster: Genre-Conditioned Audio Enhancement with Differentiable DSP

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***Abstract*—Audio mastering is inherently style dependent: the same spectral, dynamic, and loudness decisions that improve one genre may be inappropriate for another. Existing neural audio enhancement systems rarely model this dependency explicitly, and the lack of large paired mastering datasets makes direct supervised automatic mastering difficult to study. This paper presents GenreMaster, a genre-conditioned audio enhancement framework that combines Transformer-based spectral encoding, feature-wise linear modulation, learnable genre embeddings, and differentiable digital signal processing. Rather than generating audio directly, the model predicts interpretable processing parameters for a mastering-inspired effect chain. We evaluate the framework on GTZAN using controlled synthetic degradation as a proof-of-concept setting. The results show that genre-conditioned transformations can be learned end-to-end and that differentiable DSP provides a useful bridge between neural optimization and audio-engineering structure. The study does not claim production-ready mastering; instead, it identifies a practical path toward genre-aware automatic mastering once suitable paired datasets and perceptual evaluations become available.**

***Index Terms*—audio enhancement, automatic mastering, genre conditioning, differentiable digital signal processing, FiLM conditioning, music information retrieval**

I. Introduction

AUDIO mastering is the final stage of music production, where a mix is prepared for release through adjustments to loudness, spectral balance, dynamics, and stereo image. These decisions are rarely neutral. A hip-hop master may emphasize low-frequency weight and controlled impact, a classical master may preserve transient detail and dynamic range, and a rock master may favor density without sacrificing intelligibility. As a result, mastering is not only a signal enhancement problem; it is also a style-conditioned decision process. Recent progress in neural audio processing has produced strong results in enhancement, separation, restoration, and production-oriented modeling [1], [2]. However, most systems treat enhancement as a largely genre-agnostic transformation. This assumption is limiting for music production, where the desired target is shaped by genre conventions, listener expectations, and production aesthetics. A system that optimizes average reconstruction quality may still fail to make style-appropriate processing choices.

Developing automatic mastering systems also faces a data bottleneck. Supervised mastering would ideally require paired examples of unmastered and professionally mastered audio, together with genre or style annotations. Such data is difficult to obtain at scale because professional masters are proprietary and unmastered sources are seldom released alongside final versions. Consequently, research in this area must distinguish between technical feasibility and production readiness. A controlled experiment can test whether a model can learn genre-dependent transformations, but it cannot by itself establish that the model performs professional mastering.

This paper investigates genre-conditioned neural audio enhancement as a step toward automatic mastering. We introduce GenreMaster, a framework that encodes audio features with a Transformer, represents genre labels through learned embeddings, injects those embeddings using feature-wise linear modulation (FiLM), and renders the output through a differentiable digital signal processing (DSP) chain. The model therefore learns to predict parameters for interpretable audio operations rather than directly synthesizing a waveform.

To evaluate GenreMaster in a proof-of-concept setting, we utilize the GTZAN dataset to bypass the lack of large, paired mastering datasets. We construct synthetic input-target pairs by applying pre-mastering-like degradation to finished audio, then training the model to recover the original signal characteristics. This intentionally limited setup tests whether the proposed conditioning and differentiable processing architecture can learn controlled, genre-dependent transformations, rather than attempting to replace a professional mastering engineer.

The central question addressed in this paper is: can genre information improve a neural audio enhancement pipeline when the processing itself is constrained to an interpretable differentiable DSP chain? The results indicate that the proposed architecture learns stable transformations across the GTZAN genres, generalizes from validation to test data, and provides a useful experimental foundation for future work with real mastering pairs.

The main contributions of this paper are as follows:

- We present a genre-conditioned audio enhancement framework that combines Transformer spectral encoding, learned genre embeddings, FiLM conditioning, and differentiable DSP.
- We formulate mastering-inspired enhancement as interpretable parameter prediction rather than direct waveform generation.
- We evaluate the framework in a controlled GTZAN proof-of-concept setting with synthetic degradation and stratified genre splits.
- We analyze the gap between controlled reconstruction performance and production-ready automatic mastering, identifying dataset, evaluation, and style-representation requirements for future work.

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This manuscript reports implementation and experiment results from the GenreMaster project.

II. Related Work

Automatic mastering aims to approximate the final production decisions normally made by an audio engineer. Commercial services such as LANDR and eMastered indicate practical demand, but academic progress remains constrained by the scarcity of paired unmastered and mastered material. Existing research has therefore tended to approach neighboring problems, including source separation, speech and music enhancement, black-box effect modeling, and automatic mixing [3]–[6]. These areas provide useful signal-processing and learning techniques, but they do not fully address the mastering problem. In particular, most prior systems optimize enhancement or reconstruction quality without explicitly modeling the style-dependent choices that determine whether a processed track is appropriate for a given musical context.

Genre classification has long served as a central task in music information retrieval. GTZAN remains a commonly used benchmark for genre analysis despite known limitations such as fixed-length clips, coarse labels, and dataset artifacts [7]. Modern convolutional and residual encoders can learn discriminative genre representations from spectrograms, making genre labels useful control variables for downstream music-processing systems [8], [9]. However, genre labels are coarse style descriptors. They capture broad conventions, but they cannot fully represent artist identity, production era, arrangement density, recording practice, or target aesthetic. Prior genre-aware systems therefore provide a useful starting point for conditioning, while also highlighting the need to avoid treating genre labels as complete descriptions of musical style.

Feature-wise linear modulation provides a simple mechanism for injecting conditioning information into neural networks [10]. By learning scale and bias transformations from a control variable, FiLM can adapt intermediate representations without replacing the main encoder. Differentiable DSP offers a complementary direction. Instead of relying only on unconstrained waveform generation, differentiable audio effects allow gradient-based optimization over physically meaningful parameters [11], [12]. This is especially relevant for production tasks, where equalization, compression, limiting, and related effects have established engineering interpretations. Multi-resolution spectral losses further support perceptually relevant optimization in audio generation and enhancement [13], [14].

Taken together, prior work suggests three unresolved issues: mastering-oriented enhancement remains underexplored relative to adjacent audio tasks, genre information is useful but incomplete as a style representation, and interpretable differentiable processing has not been sufficiently studied in genre-conditioned mastering settings. The present study addresses these gaps by evaluating style-conditioned enhancement under a controlled protocol, using genre information as an explicit conditioning signal while keeping the learned transformation grounded in differentiable audio-processing operations.

III. Methodology

A. Problem Formulation

Let x denote an input audio signal and let g denote its genre label. The objective is to learn a transformation f_θ that produces an enhanced signal $\hat{y}$:

$$\hat{y} = f_\theta(x, g). \tag{1}$$

In GenreMaster, this transformation is decomposed into audio encoding, genre conditioning, DSP-parameter prediction, and differentiable rendering. Rather than modeling $\hat{y}$ as an unconstrained neural waveform, the system predicts a parameter vector ϕ for a mastering-inspired processing chain:

$$\phi = P_\theta(E_\theta(x), z_g), \tag{2}$$

$$\hat{y} = D_\phi(x), \tag{3}$$

where E_θ is the audio encoder, z_g is the learned genre embedding, P_θ is the FiLM-conditioned parameter predictor, and D_ϕ is the differentiable DSP chain.

The parameter vector is structured around the mastering controls predicted by the model:

$$\phi = [\phi_{\text{lufs}}, \phi_{\text{eq}}, \phi_{\text{comp}}, \phi_{\text{lim}}, \phi_{\text{width}}], \tag{4}$$

where the terms denote target loudness, equalization, multiband compression, limiter threshold, and stereo-width control, respectively.

This formulation is designed to preserve a connection between the learned model and audio-engineering practice. The network learns how strongly to apply processing, while the rendering stage remains structured around recognizable audio operations.

B. Architecture

The architecture consists of four components, shown in Fig. 1. First, log-mel spectral features are extracted from the input audio and passed to a Transformer encoder [15]. The encoder models temporal dependencies in the spectral representation and produces a compact audio feature vector.

Second, each genre label is mapped to a learned embedding. This embedding represents genre as a continuous conditioning signal rather than as a hard-coded rule set. Third, the genre embedding modulates the audio representation using FiLM:

$$\text{FiLM}(h, z_g) = \gamma(z_g) \odot h + \beta(z_g), \tag{5}$$

where h is the encoded audio representation and $\gamma(\cdot)$ and $\beta(\cdot)$ are learned affine transformations of the genre embedding.

Finally, the conditioned representation is mapped to parameters for a differentiable DSP chain. The chain includes loudness control, parametric equalization, multiband compression, limiting, and stereo-width adjustment. These modules are selected because they correspond to common mastering operations while remaining amenable to differentiable optimization.

Training minimizes a weighted combination of loudness, spectral, dynamic, and perceptual terms:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{loud}}\mathcal{L}_{\text{loud}} + \lambda_{\text{spec}}\mathcal{L}_{\text{spec}} + \lambda_{\text{dyn}}\mathcal{L}_{\text{dyn}} + \lambda_{\text{perc}}\mathcal{L}_{\text{perc}}. \tag{6}$$

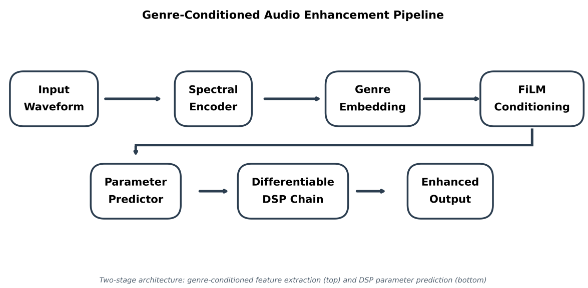

Fig. 1. GenreMaster pipeline. Audio features are encoded, modulated by learned genre embeddings, and mapped to parameters for a differentiable mastering-inspired DSP chain.

Two components are particularly important for mastering-oriented behavior. The loudness term penalizes differences in global level using an RMS-based decibel approximation:

$$\mathcal{L}_{\text{loud}} = \frac{1}{B}\sum_{i=1}^{B} \left|20\log_{10}(\text{RMS}(\hat{y}_i)+\epsilon) - 20\log_{10}(\text{RMS}(y_i)+\epsilon)\right|, \quad (7)$$

where B is the batch size and ϵ is a small constant for numerical stability. The dynamic term penalizes differences in crest factor so that peak-to-average behavior is also matched:

$$\mathcal{L}_{\text{dyn}} = \frac{1}{B}\sum_{i=1}^{B} \left|20\log_{10}\frac{p(\hat{y}_i)+\epsilon}{\text{RMS}(\hat{y}_i)+\epsilon} - 20\log_{10}\frac{p(y_i)+\epsilon}{\text{RMS}(y_i)+\epsilon}\right|, \quad (8)$$

where $p(\cdot)$ denotes a differentiable peak approximation. The loss design reflects the fact that mastering quality cannot be captured by a single waveform distance. Loudness alignment, spectral balance, dynamic behavior, and perceptual similarity each capture a different aspect of the desired transformation. In the present proof-of-concept, the loss is used to test learnability under controlled conditions rather than to define a complete perceptual mastering criterion.

C. Data and Experimental Setup

The experiment uses GTZAN after removing one corrupted file, yielding 999 valid clips across ten genres. The data are split in a stratified manner into training, validation, and test partitions, as summarized in Table I. All audio is resampled to 22,050 Hz and represented as 30-second clips. Since GTZAN contains finished music rather than unmastered/mastered pairs, we construct synthetic training pairs to evaluate the model. We create the input by applying degradation operations that imitate common pre-mastering deficiencies. Using the original clip as a pseudo-target provides a controlled learning problem, though it should not be interpreted as a faithful simulation of professional mastering.

To establish that the GTZAN labels provide a meaningful conditioning signal, we first trained a separate ResNet-based genre classifier on the same corpus. This auxiliary stage was used to verify that genre structure is discriminable in the dataset before running the mastering experiment. The reported Transformer-based mastering model was then trained with genre labels as the conditioning variable, using learned genre embeddings within the FiLM-conditioned parameter-prediction pipeline.

For the mastering model, training was configured for a maximum of 50 epochs with validation monitored after each epoch. Gradient accumulation was used to increase the effective batch size, the learning rate followed a cosine schedule with a short warmup phase, and the best checkpoint was selected according to validation loss. Early stopping was enabled with a patience of 15 epochs without improvement, while intermediate checkpoints were saved periodically for recovery and inspection.

TABLE I
DATASET SPLIT CONFIGURATION

Set	Clips
Training	799
Validation	99
Test	101

TABLE II
TRAINING CONFIGURATION

Component	Value
Random seed	42
Sample rate	22,050 Hz
Clip duration	30.0 s
Batch size	8
Gradient accumulation	4
Effective batch size	32
Maximum epochs	50
Optimizer	AdamW [16]
AdamW betas	$(0.9, 0.999)$
AdamW epsilon	1×10^{-8}
Learning rate	3×10^{-4}
Weight decay	1×10^{-4}
Scheduler	Cosine with warmup
Warmup epochs	2
Gradient clipping	1.0
Checkpoint interval	Every 5 epochs
Model selection	Lowest validation loss
Early stopping	Patience 15
Executed epochs	32
Device backend	CUDA

IV. RESULTS

A. Training Behavior

Fig. 2 shows the training and validation trajectories. The model converges rapidly and maintains a small gap between training and validation loss. This behavior suggests that the architecture is able to learn the controlled degradation-reversal task without severe overfitting.

B. Test Performance

Table III reports the held-out test results. The test loss closely matches the best validation loss, and waveform-level metrics indicate strong reconstruction quality in the synthetic setting. These values should be interpreted as evidence that the architecture learns the controlled transformation, not as evidence of professional mastering quality.

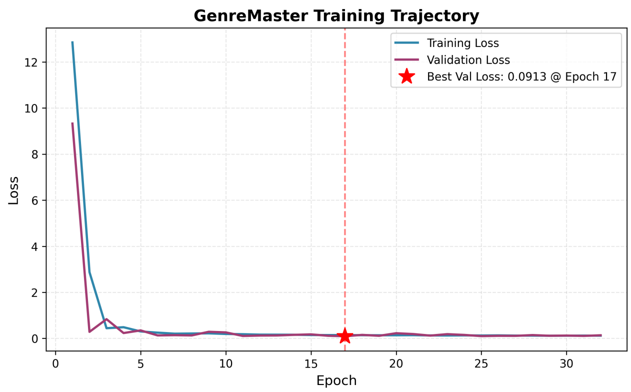

Fig. 2. Training and validation loss curves for GenreMaster on the controlled GTZAN enhancement task.

TABLE III
HELD-OUT TEST PERFORMANCE

Metric	Value
Test samples	101
Validation loss	0.0913
Test loss	0.0903
SNR	16.67 dB
Correlation	0.9892
Best epoch	17

C. Genre-Dependent Behavior

Fig. 3 reports test loss by genre. The losses remain within a narrow range across the ten GTZAN categories. This consistency suggests that genre conditioning does not simply benefit one or two genres while failing on others. Instead, the framework learns a shared enhancement structure that can be adapted through the genre embedding.

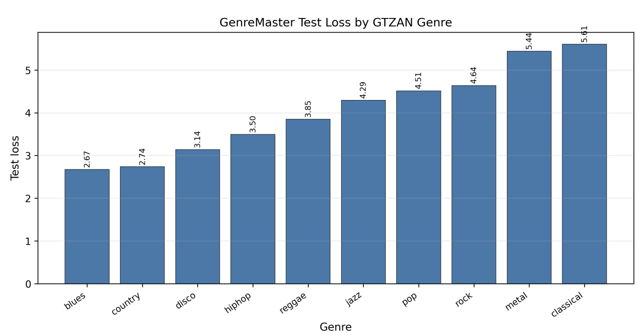

Fig. 3. Per-genre test loss on GTZAN. The narrow spread indicates stable behavior across the evaluated genre labels.

D. Auxiliary Genre Separability

To verify that the dataset supports meaningful genre-conditioned modeling, we first trained a separate ResNet-based genre classifier on the same corpus. The classifier reached high validation accuracy, as shown in Fig. 4, indicating that the genre labels are learnable from the spectral representation. This auxiliary result motivated the subsequent mastering experiment, in which genre labels were used as the conditioning signal for the Transformer-based model. At the same time, GTZAN genre labels remain coarse and should not be treated as complete style descriptors.

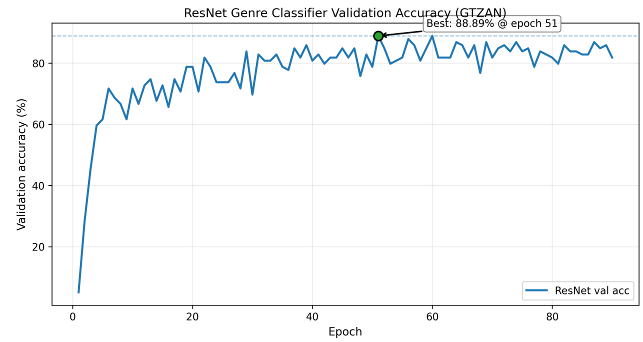

Fig. 4. Validation accuracy trajectory for the auxiliary ResNet genre classifier on GTZAN.

V. DISCUSSION

The results show that a neural system can learn genre-conditioned audio transformations when the output space is constrained through differentiable DSP. The small validation-test gap indicates stable generalization within the controlled GTZAN setting. The per-genre analysis further suggests that the model does not rely on a single global correction but adapts its processing across genre labels. The architecture also provides a practical advantage over direct waveform generation. Because the model predicts parameters for familiar audio effects, its behavior can be inspected in terms closer to mastering practice. This does not make the system equivalent to a mastering engineer, but it gives the learning problem a useful structure and reduces the opacity of the output transformation.

The main limitation is the absence of real mastering pairs. GTZAN contains already-finished music, and the degraded inputs are generated synthetically. Real pre-mastering material differs in recording quality, mix balance, headroom, noise, room coloration, arrangement density, and artistic intent. A model trained only to reverse synthetic degradation may learn useful signal transformations without learning the judgment required for professional mastering.

The evaluation metrics are also limited. Reconstruction loss, correlation, and SNR are useful for controlled experiments, but mastering is ultimately perceptual and contextual. A technically close reconstruction is not necessarily a better master. Future evaluations should include loudness and true-peak compliance [17], perceptual audio metrics [18], [19], comparisons with commercial systems, and listening studies involving trained engineers.

Finally, genre labels are an imperfect representation of style. Production conventions vary within genres and evolve over time. A more complete system should learn continuous style embeddings that can capture subgenre, era, artist preference, and reference-track intent.

A. Path Toward Automatic Mastering

The present study suggests three requirements for moving from proof-of-concept enhancement to automatic mastering. First, the field needs paired datasets containing unmastered and professionally mastered versions across diverse genres and production contexts. Second, models should be evaluated with perceptual and preference-based criteria rather than reconstruction metrics alone. Third, conditioning should move beyond discrete genre labels toward richer style representations that can reflect real mastering goals.

Under these conditions, the GenreMaster architecture could serve as a foundation for a more realistic system: the Transformer encoder would model musical context, the conditioning pathway would represent target style, and the differentiable DSP chain would provide controllable, interpretable rendering.

VI. Conclusion

This paper presented GenreMaster, a genre-conditioned audio enhancement framework that combines Transformer spectral encoding, FiLM conditioning, learned genre embeddings, and differentiable DSP. The model predicts interpretable parameters for a mastering-inspired processing chain rather than generating audio directly.

In a controlled GTZAN proof-of-concept experiment, GenreMaster learned stable degradation-reversal transformations and generalized consistently across held-out clips and genre categories. These findings support the feasibility of genre-conditioned differentiable DSP for music enhancement.

At the same time, the study is intentionally bounded. The experiment does not establish production-ready automatic mastering because it relies on synthetic degradation and finished music rather than true unmastered/mastered pairs. Future work should focus on real mastering datasets, perceptual evaluation, richer style conditioning, and comparisons with professional and commercial mastering systems.

Ethics and Reproducibility Statement

All experiments used publicly available music data and code-executed metrics from repository artifacts. No human subject data were collected. The reported values are derived from saved experimental outputs, and the limitations of the proof-of-concept evaluation are stated explicitly.

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